

Demonstrating Augmented Breathing via Thermal Feedback in the Nose

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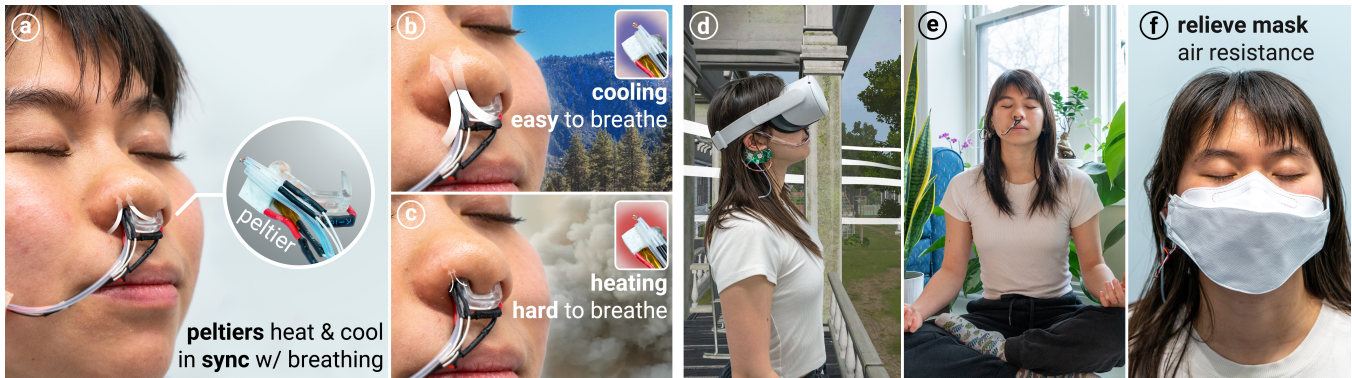


Figure 1: (a) We demonstrate a novel nasal interface that augments the perception of breathing via thermal feedback in the nose. By cooling or heating the nasal cavity in sync with inhalations, our device can make the user feel as if they are (b) breathing easier (i.e., inhaling more air) by cooling the inside of the nose or (c) having a harder time breathing (i.e., stuffy room or blocked nose) by heating. Based on study participants’ analogies, we demonstrate our method with several applications: (d) a VR game that simulates experiences like breathing dust or fresh air in the outdoors; (e) a meditation app that draws the user’s attention to their breathing; and (f) an application that relieves mask-wearers from the perceived air resistance while breathing through the mask filter.

ABSTRACT

We demonstrate a novel method to augment the feeling of breathing—enabling interactive applications to let users feel like they are inhaling more/less air (perceived nasal airflow). We achieve this effect by cooling or heating the nose in sync with the user’s inhalation. Our illusion builds on the physiology of breathing: we perceive our breath predominantly through the cooling of our nasal cavities during inhalation. This is why breathing in a “fresh” cold environment feels easier than in a “stuffy” hot environment, even when the inhaled volume is the same. Following, we engineered a compact device worn across the septum that uses Peltier elements. We illustrate the potential of this augmented breathing in a demonstration with two tracks: a fast track for a quick experience and a slow track

for an immersive virtual reality (VR) walkthrough. In the fast track, visitors can quickly experience altered breathing perception using a simple application and our devices. In the slow track, visitors can put on a VR headset and experience an interactive scene where their sense of breathing is altered (e.g., rendering ease of breathing crisp air or difficulty breathing with a deteriorated gas mask).

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices**; • **Hardware** → **Emerging interfaces**.

KEYWORDS

Breathing, Thermal, Trigeminal, Perception, Respiration

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1 INTRODUCTION

Breathing stands out as a unique autonomic function because it is primarily controlled by the brain [10, 11], resulting in a significant impact on human experiences, particularly our comfort and emotional state [1, 9]. Additionally, as it is commonly practiced in everyday life, voluntary deep breathing affects the autonomic nervous system, modulating comfort and stress levels [2].

As such, it is unsurprising that breath-responsive interactive systems and experiences are increasingly popular in human-computer interaction. For instance, a number of games have integrated breath as one of their input modalities [6, 12, 13, 15, 18]. Further, implementing breath as an input has been a common feature in biofeedback systems to support calming interventions, such as promoting stress relief through visual or auditory feedback [5, 7, 8, 16].

While leveraging the user’s breathing as an input channel has been explored and can be implemented via a wide range of solutions (e.g., hot-wire [17] or -film [3] sensing, microphones [18], force sensors on the chest [13]), the reverse is not the same—output devices that can alter the user’s breathing are very limited.

Output devices for breathing remain non-trivial since, to alter one’s breathing, the user’s airflow needs to be altered. In fact, interactive devices that can achieve this are extremely rare. One such example is *AirRes Mask*, a mask-based device that uses a servo motor to block the entrance of a tube the user breathes through, thereby physically restricting airflow to enhance virtual reality immersion [14]. This unique example highlights the challenges in engineering a suitable method for altering breath, as to manipulate inhaled airflow, one requires motors, tubes, pumps, and more, making these devices cumbersome and non-practical, which might explain why so few such devices have been proposed.

In this paper, we demonstrate a different approach to altering the user’s airflow, one that does not require large tubes, pumps, and other mechanical actuators—we alter the user’s perceived airflow by cooling down or heating up their nose using Peltier devices, as shown in Figure 1. Our illusion builds on the physiology of breathing: we perceive our breath predominantly through the cooling of our nasal cavities during inhalation. Our user study confirmed that this type of in-nose temperature stimulation significantly influenced breathing perception in both directions: making it feel harder or easier to breathe.

To enable this novel sensation in interactive applications, we engineered a compact device worn across the septum featuring Peltiers and air pressure sensors. Finally, we illustrate the potential of augmented breathing for a range of contexts, from virtual reality to everyday interactions.

2 DEMONSTRATION

To illustrate our technique with an application, we designed a demonstration set-up for ACM UIST 2024. Our demo will feature two tracks: a fast track and a slow track, catering to different levels of visitor engagement.

2.1 Fast track: Mini-booths

The fast track is designed for visitors who do not want to wait for the slow track (virtual reality experience). Instead, they can come to

a table at the booth with two computers and two augmented breathing devices. The computers will display their breathing patterns and allow them to activate either increased or decreased perceived airflow. Visitors can choose to either watch or listen to a matching scene, such as seeing mountains and perceiving an increase in airflow or watching traffic and perceiving a decrease. This setup is intended to provide a quicker, yet still engaging, experience of our augmented breathing technology.

2.2 Slow track: VR experience

The slow track is an adaptation of our paper’s virtual reality (VR) walkthrough demonstration, focusing on 1-2 interactions using augmented breathing in VR. Visitors to the demo booth can put on the VR headset and the device, and experience an interactive scene. This immersive setup will showcase how our wireless wearable device, equipped with Peltier elements and airflow sensors, enables real-time adjustments to the temperature inside the user’s nose, altering their breathing perception. Drawing on insights from our user study, we highlight direct quotes and real-world analogies participants used to describe augmented breathing. These highlight how participants perceived changes in ease of breathing, such as feeling like breathing “more air” or inhaling “through a narrow straw.” These augmentations are purely perceptual; the device does not physically restrict airflow, allowing users constant access to air while experiencing a sensation of altered nasal airflow. Below, we reproduce the full walkthrough, though visitors will experience a condensed version.

Figure 2 depicts our VR user trying to escape a room where they were being held. As they walk through this old house in search of the door’s key, their movements kick dust into the air. In sync with their breathing, our device heats the inside of their nose, creating a brief sense of breathing less due to the dusty and “stuffy room” (quote from Study, by P1).

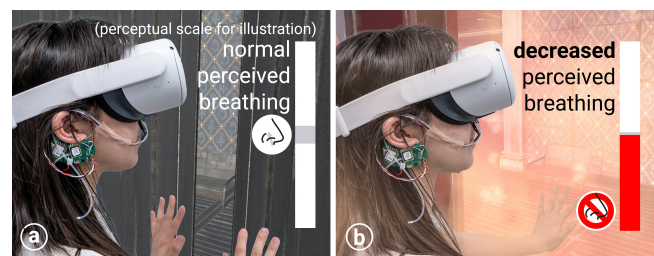


Figure 2: (a) The user opens a room and (b) kicks up dust. Scale depicts perceived nasal airflow for illustration purposes only.

As depicted in Figure 3, upon finding the key and opening this room’s door, the user inadvertently triggers the alarm, releasing smoke—in sync with inhaling, our device increases the nose’s temperature to emphasize the feeling of “*smog in the air*” (direct quote from Study, by P8).

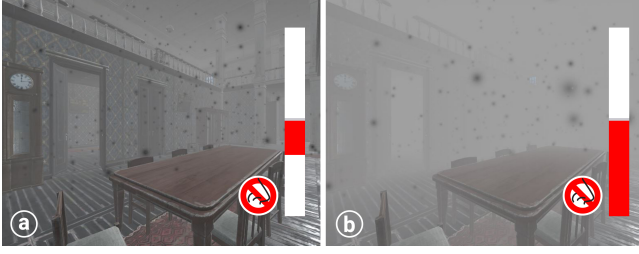


Figure 3: (a) Smoke starts to fill the house with black specks and smog until (b) it feels as if it becomes harder to breathe.

Reacting to the smog, shown in Figure 4, the user finds a gas mask and equips it. Immediately, our device starts to cool off their next inhalations, creating a feeling of “more air entering my nostrils” (direct quote from Study, by P6).

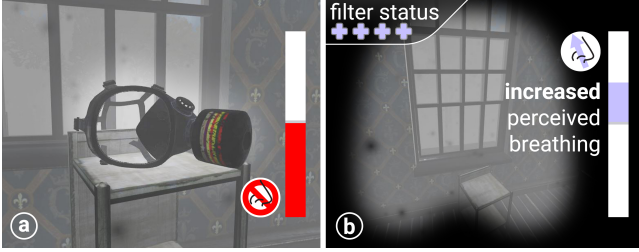


Figure 4: (a) User finds a mask, and (b) equips it, which causes our device to increase their feeling of perceived nasal airflow.

However, as the virtual mask’s filter deteriorates, the perceived breathing becomes harder, which can be felt as “a lot more effort to move air into my lungs” (direct quote from Study, by P4). As shown in Figure 5, to render this sensation, our device progressively increases the temperature during each inhalation (i.e., fading from cold to hot). As the user finds a new mask and equips it, our device responds by cooling their inhalations and allowing the user to feel like breathing “more air” (direct quote from Study, by P3).

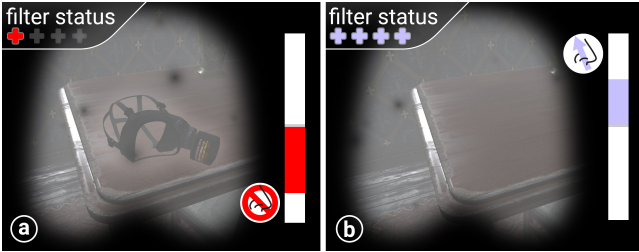


Figure 5: (a) As the mask’s filter deteriorates, our device renders a difficulty in inhaling air. (b) With a replacement, our device resets the perceived nasal airflow sensation.

Finally, the user opens the door of the gas-filled room and escapes. As they open the doors (Figure 6), our device pushes through

its maximum cooling effect to emphasize the transition from claustrophobic to the relief of “breathing in an open space” (direct quote from Study, by P2).

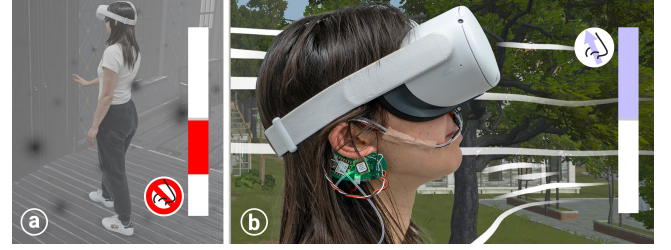


Figure 6: (a) The user opens the final door, leading to an (b) open space where they feel as if they breathe more easily.

3 IMPLEMENTATION

To help readers replicate our design, we provide technical details and an open-source repository.¹ Figure 7 shows our device worn on the nose and around the user’s ear.

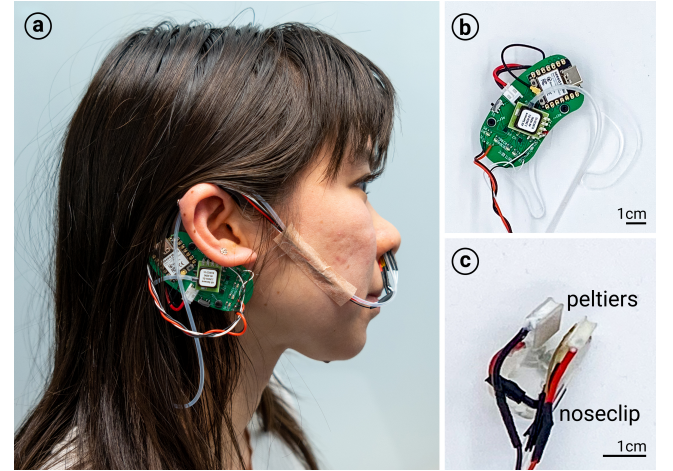


Figure 7: (a) A user wearing our device around the ear and in the nose, comprised of its (b) control unit and (c) nose clip.

Circuit design. Our circuit, depicted in Figure 8, was engineered around an ESP32C3 with Bluetooth Low Energy. To change the nose’s temperature, we attached two 8 mm × 8 mm Peltiers (*Custom Thermoelectric*, 00711-5L31-03CA-S) paired with thin 10 kΩ thermistors (*TME*, B3950) for temperature monitoring. We also added a 10 kΩ thermistor and a small silicone tube (1 mm ID, 2 mm OD) in one nostril to record the baseline temperature during inhalation, providing a reference point for temperature adjustments. This silicone tube connected to a differential air pressure sensor (*Amphenol All Sensors*, 1 INCH-D1-4V-MINI) within an operating range of −250 Pa to 250 Pa to monitor respiration. Finally, we concentrated most of the components behind the ear (microcontroller, battery,

¹<https://github.com/humancomputerintegration/augmented-breathing>

etc.). The in-nose components weighed only 7 g, while the components positioned behind the ear weighed 26 g. This separation balanced functionality with wearability.

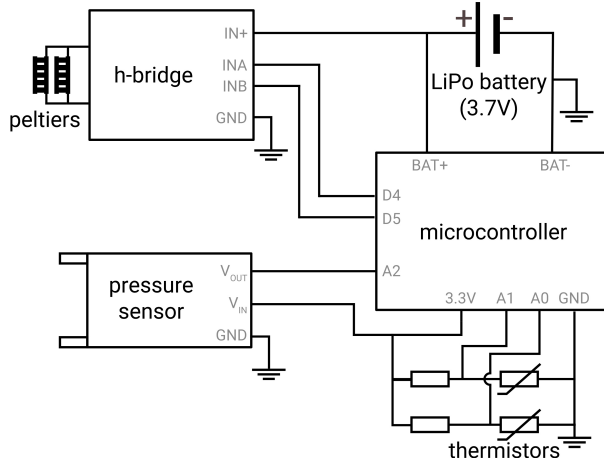


Figure 8: High-level schematic of our device's circuitry.

Control loop. To regulate the temperature of the Peltiers, we tuned a PID controller, which activates if the temperature is more than 0.25°C from the set point. The activation of the Peltiers is synchronized with the user's inhalation, capitalizing on the exhalation phase as a natural cooling period that allows the Peltiers to return to the user's baseline body temperature, thus conserving power without the need for active cooling.

Inhalation detection. To achieve accurate inhalation detection, our system includes a calibration process. Initially, users hold their breath for two seconds to establish a baseline for the differential air pressure (difference between the pressure measured in the nostril and pressure measured in the room) while not breathing, representing the static noise level of air pressure. The device then records the mean and standard deviation of this baseline pressure. An inhalation is detected when the differential air pressure falls two standard deviations below this mean. Likewise, an exhalation is identified when it rises two standard deviations above.

Latency. To assess system latency, specifically the delay between the temperature peak induced by our device and the peak of inhalation, we analyzed recorded data from all participants. This dataset comprised 3247 breaths, each with temperature stimulation. We found an average latency of 0.21 ± 0.67 s between the onset of the temperature peak (when the Peltier reaches its extreme temperature) and the peak of inhalation (the point of maximum inhalation). Considering that the average inhalation duration was 1.81 ± 0.49 s, a latency of 0.21 s was reasonable for our interactive system. Notably, the temperature peak occurred within the inhalation phase for 91.32 % of the breaths, confirming effective synchronization with the breathing cycle.

Power consumption. Our prototype drew a maximum of 6 W during inhalation. Under a continuous change of perceived airflow only during inhalation, it lasted approximately 22 min with a small 300 mA h Li-Po battery. The device could operate longer with larger batteries, but we noted that 22 min of perceived airflow change was

relatively long, considering participants in our studies detected a change in perceived airflow within their first inhalation with our system active.

4 CONCLUSIONS

We demonstrated a technique for altering the perception of breathing through thermal feedback, using Peltiers to alter the user's sensation of nasal airflow. Unlike existing systems that rely on mechanical parts to restrict breathing, our perceptual approach modulates airflow *perception* and not the actual volume of air that users are inhaling—making it safer than previous devices. Moreover, it is the first approach that enables *both* decreased and increased perceived airflow.

Our effect can be achieved with small Peltiers and minimal temperature adjustments. This opens potential practical implications for our findings, providing a working method to enrich interactive experiences like virtual reality and apply augmented breathing to everyday contexts.

We envision our work as a foundation for future investigations on how thermal feedback can be harnessed to augment breath perception. Future research may explore altering not only the perception of our inhalation but also our exhalation, developing power-saving strategies, exploring new actuators (e.g., flexible Peltiers [4]), or even investigating its potential impact for therapeutic purposes.

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